Title: Diabetes Prediction using Machine Learning

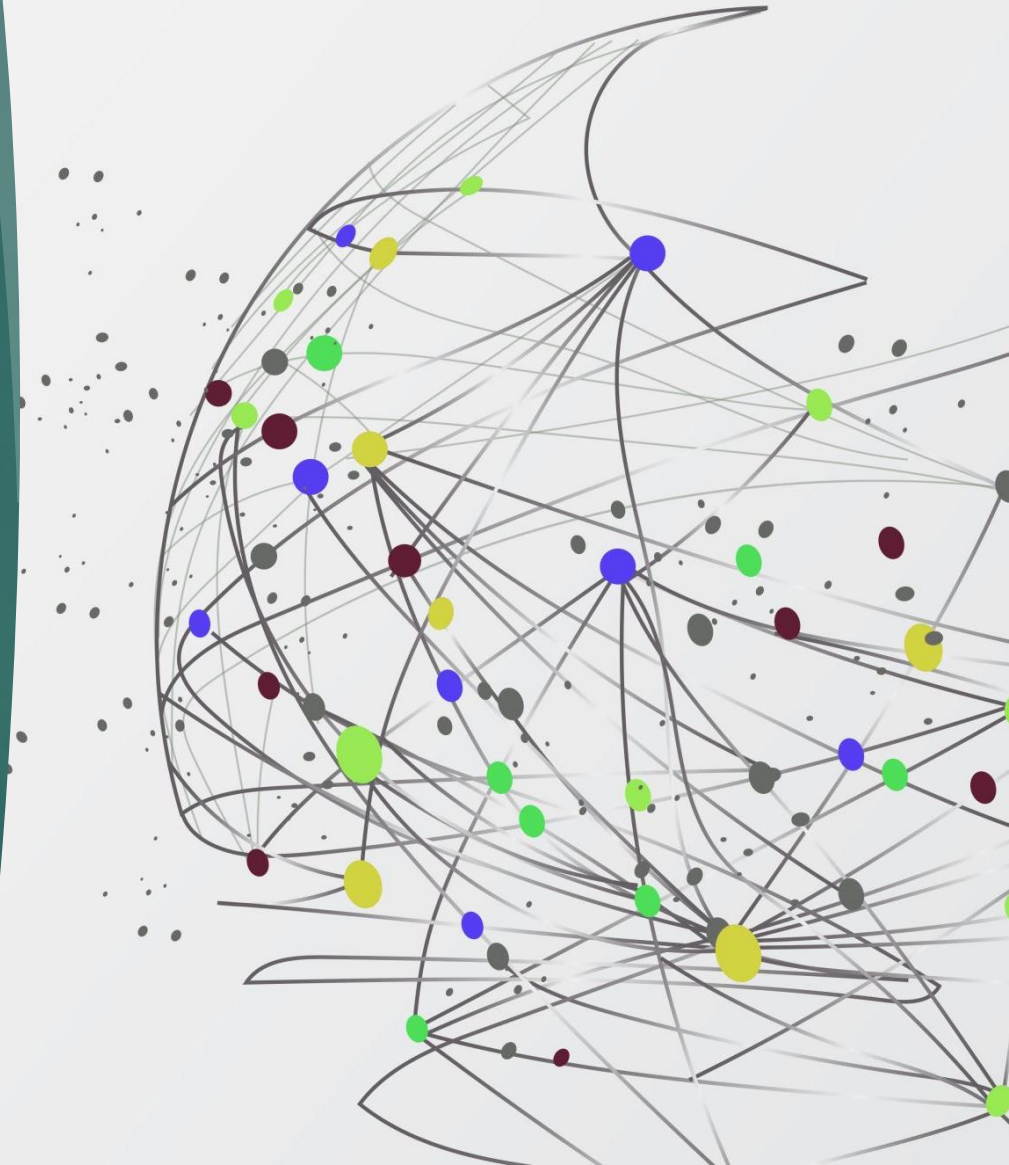
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Introduction

- Diabetes is a chronic disease affecting millions worldwide
- Early detection is very important
- Machine Learning helps predict diseases using data



Objective



PREDICT DIABETES USING PATIENT MEDICAL DATA



COMPARE DIFFERENT ML ALGORITHMS



IDENTIFY IMPORTANT HEALTH FACTORS

Dataset

- Dataset: **Pima Indians Diabetes Dataset**
- Total records: 768
- Features:
  - Glucose
  - Blood Pressure
  - BMI
  - Insulin
  - Age
- Target:
  - 0 → No Diabetes
  - 1 → Diabetes





Exploratory Data Analysis (EDA)

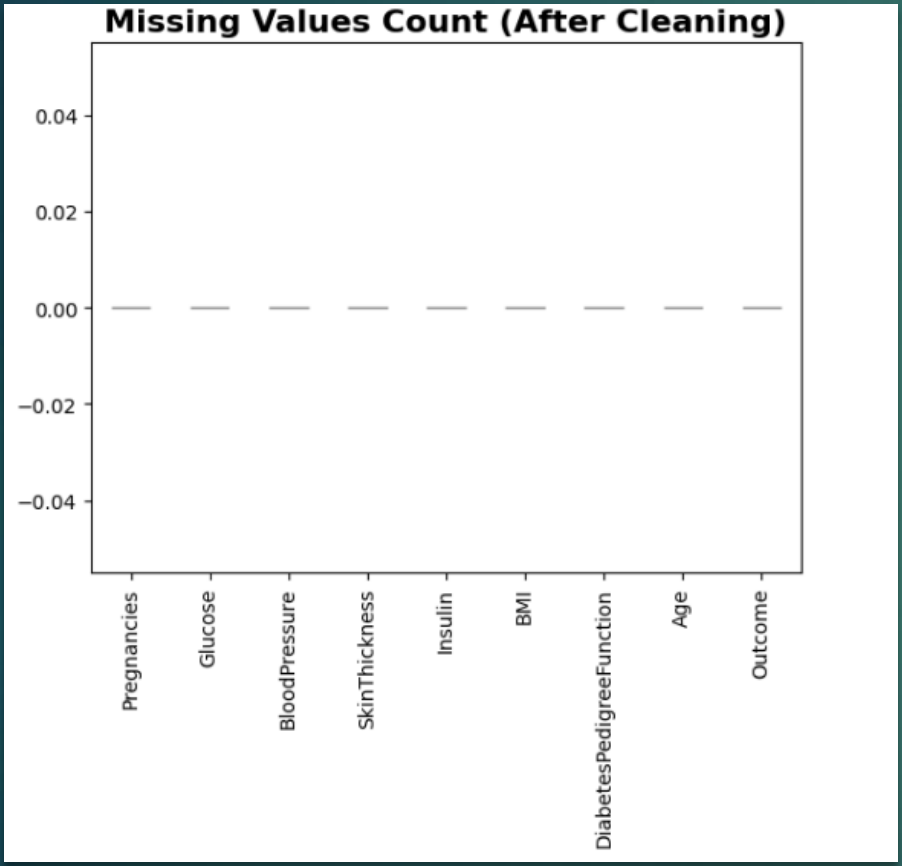
. **Missing Values Graph:**

To check whether the dataset has any missing or invalid values.

. **Correlation heatmap**

To show the relationship between different features in the dataset

Missing Values Graph



In this project:

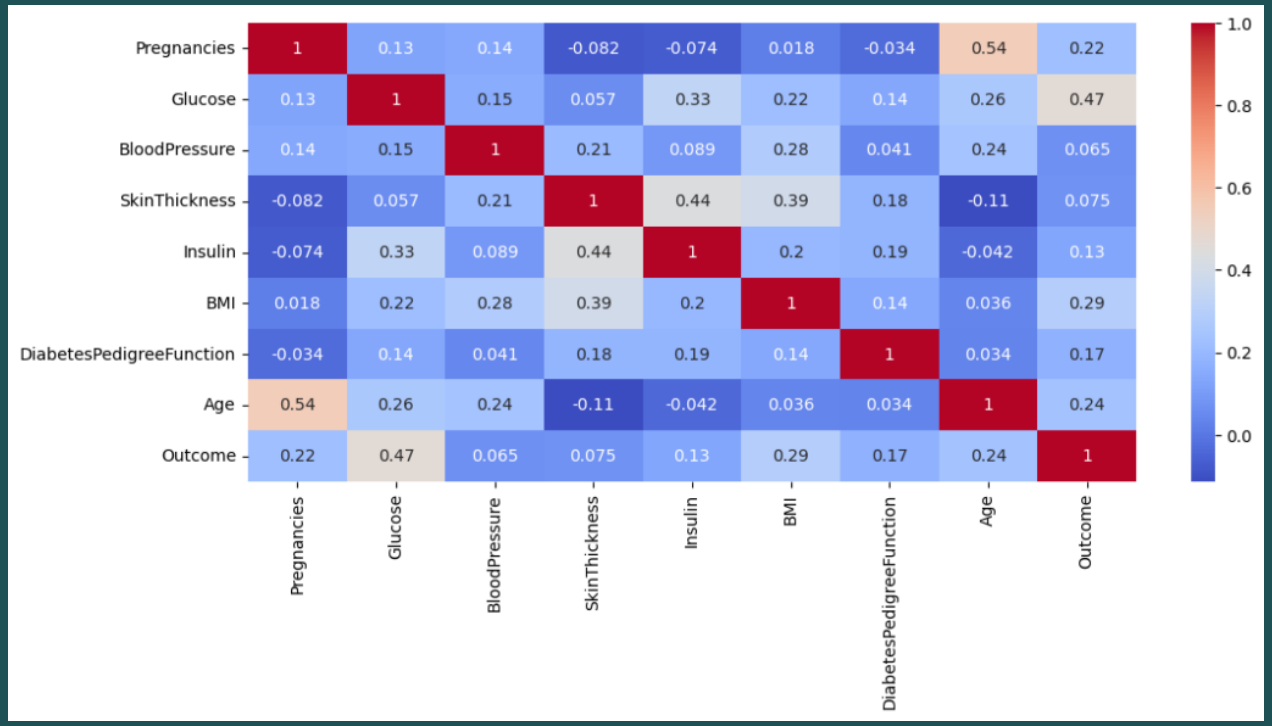
- Some columns had **0 values treated as missing**
- We replaced them with **NaN**
- Then filled them using **median**

Final result:

After cleaning, the graph shows **zero missing values**

Correlation Heatmap

- 🔴 **Red (close to +1)** → Strong positive correlation
- 🔵 **Blue (close to 0 or negative)** → Weak or no correlation
- Diagonal (1.0) → Same feature compared with itself



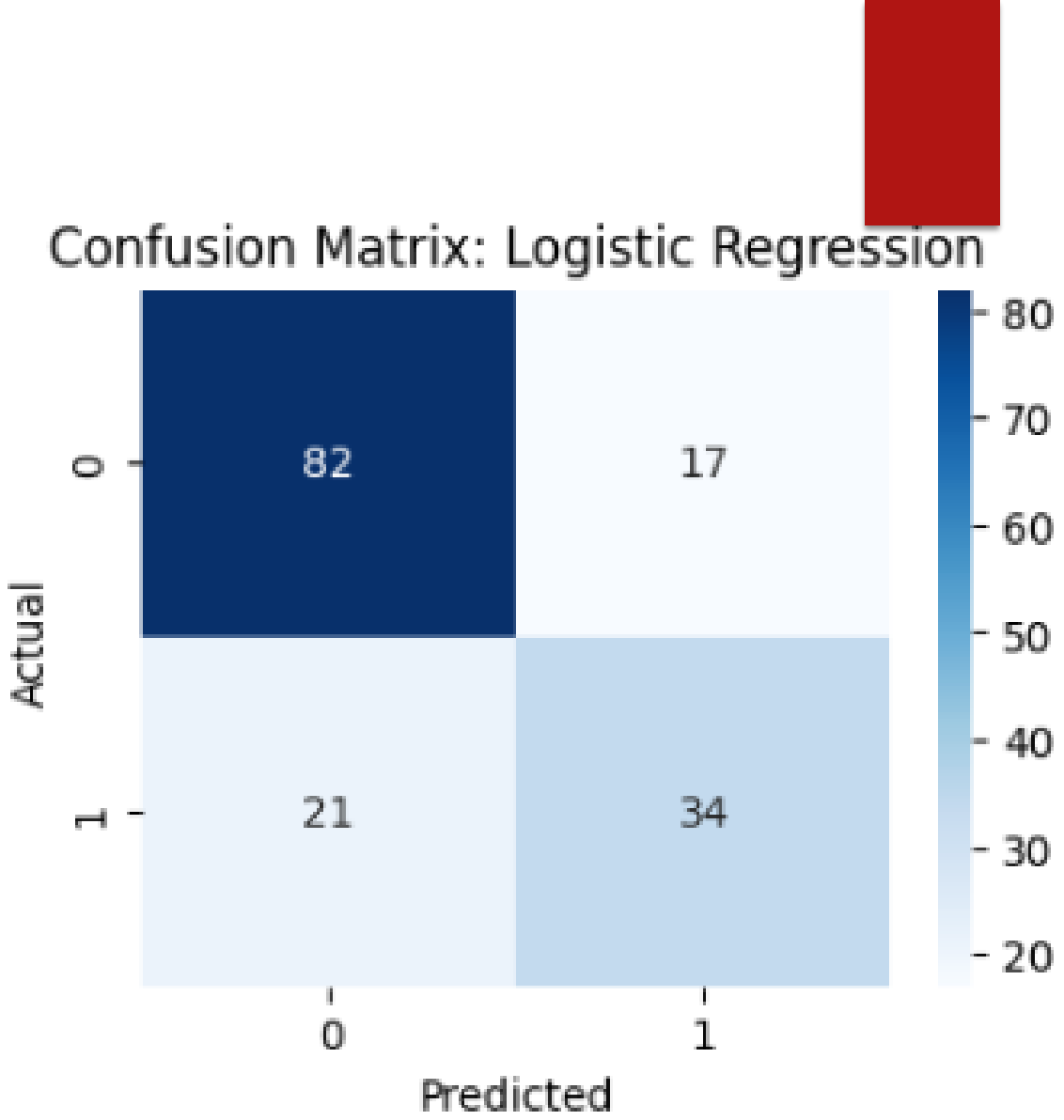
Models Used

- LOGISTIC REGRESSION
- K-NEAREST NEIGHBORS (KNN)
- DECISION TREE
- RANDOM FOREST

Confusion Matrix: Logistic Regression

- Logistic Regression is a **classification algorithm** used to predict **binary outcomes**

--- Logistic Regression Results ---
Accuracy: 0.75

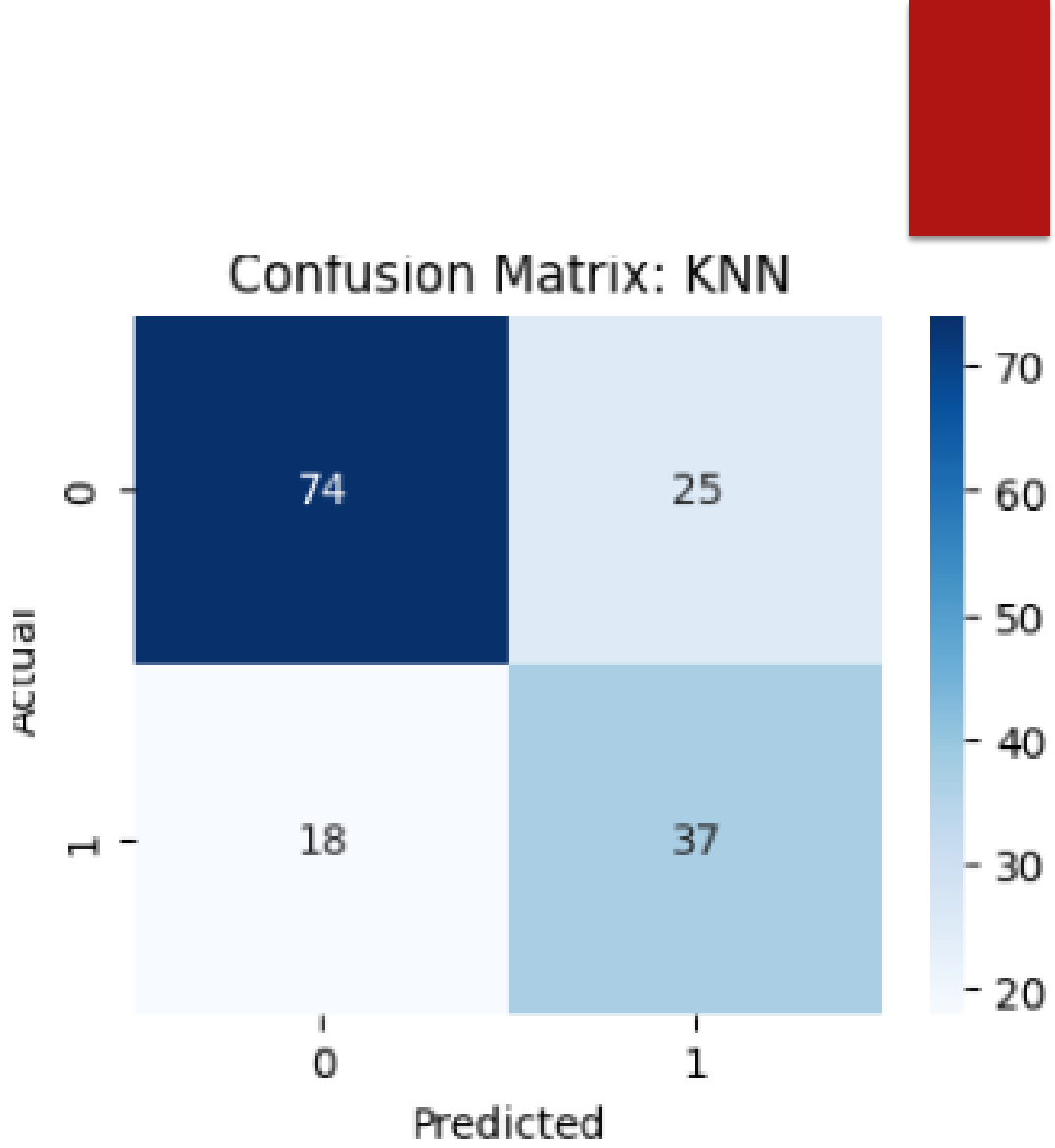


K-Nearest Neighbors (KNN) :

- KNN (K-Nearest Neighbors) is a **classification algorithm** that predicts a class based on the **nearest data points**

--- KNN Results ---

Accuracy: 0.72

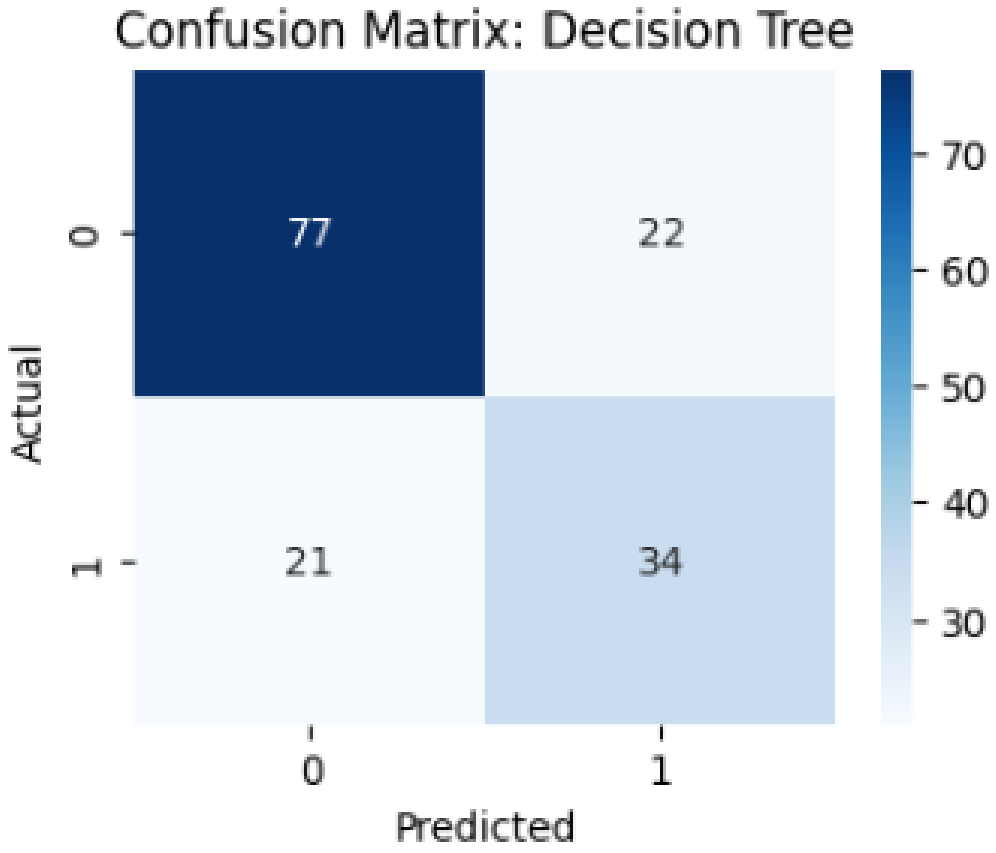


Decision Tree:

- A **Decision Tree** is a supervised machine learning algorithm used for **classification** and **regression**. It works like a flowchart where each internal node represents a decision, and each leaf node represents an output.

--- Decision Tree Results ---

Accuracy: 0.72

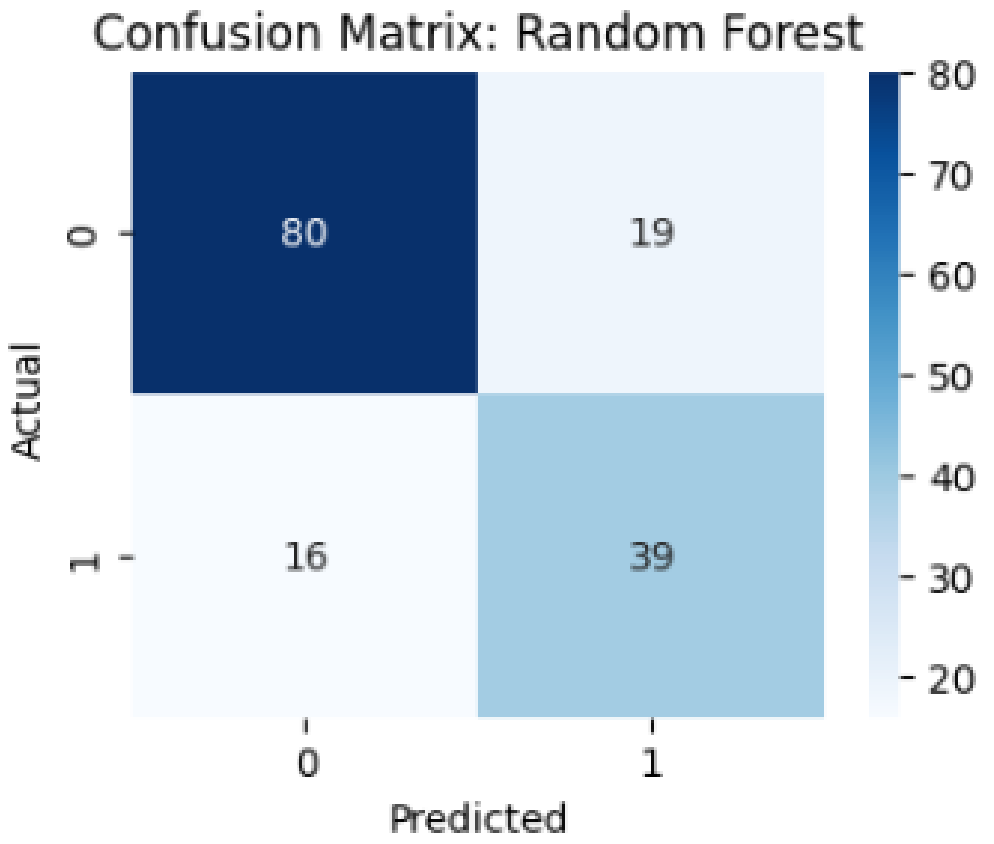


Random Forest :

- A **Random Forest** is a powerful **ensemble learning algorithm** used for **classification and regression**. It works by combining multiple **decision trees** to produce a more accurate and stable result.

--- Random Forest Results---

Accuracy: 0.77

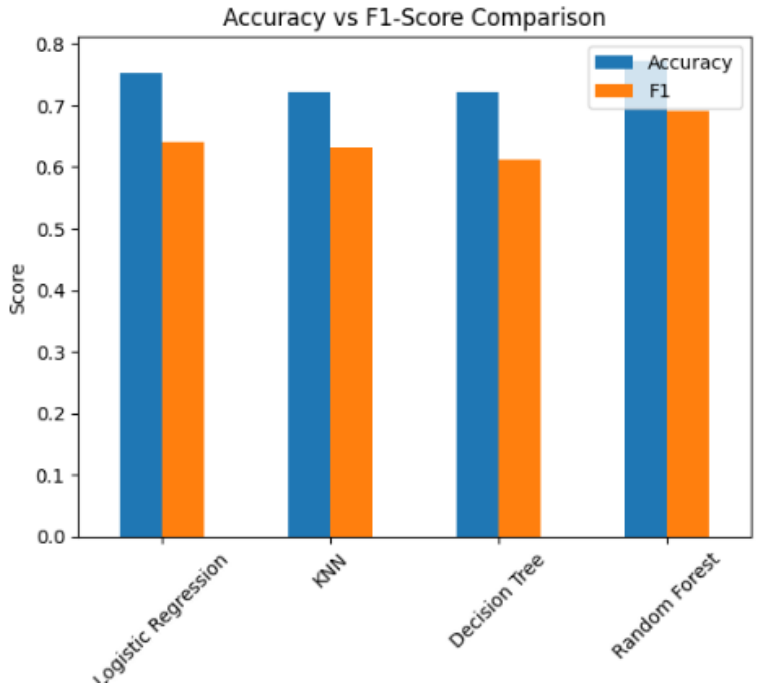


Model Comparison Table:

	Model	Accuracy	Precision	Recall	F1
0	Logistic Regression	0.753247	0.666667	0.618182	0.641509
1	KNN	0.720779	0.596774	0.672727	0.632479
2	Decision Tree	0.720779	0.607143	0.618182	0.612613
3	Random Forest	0.772727	0.672414	0.709091	0.690265

Best Model

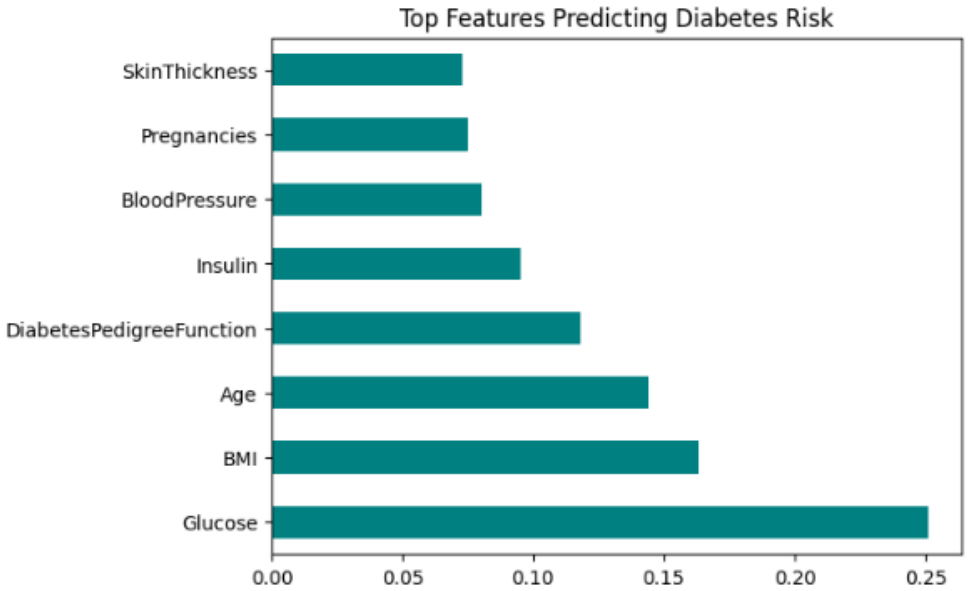
- Random Forest (usually best)
- Highest accuracy



Feature Importance

Top features:

- Glucose
- BMI
- Age



Conclusion

- Random Forest gives best accuracy
- Data preprocessing improved results
- ML can assist doctors in early detection

THANK YOU